

Rene Alzina

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EDUCATION

University of Notre Dame | Notre Dame, IN May 2027
Bachelor of Science in Engineering GPA: 3.99
Major: Computer Science | Minor: Engineering Corporate Practice (Business Management & Engineering Leadership)
Engineering Summer Abroad | Dublin, Ireland May 2025 – June 2025

TECHNICAL AND LANGUAGE SKILLS

Technical: Python, C/C++, Java, Kotlin, HTML/CSS/JS, bash, React, Pandas, Tableau, Github, Harness, Splunk
Coursework: Data Structures, Algorithms, Operating Systems, Distributed Systems, Theory of Computing, Intro to AI
Spoken Languages: Fluent in Spanish, Intermediate French

EXPERIENCE

Wells Fargo | Phoenix, AZ May 2026 – August 2026
Technology Intern

- Innovated and developed a deployment configuration tool using **Python** and **Flask** that scrapes, tabulates, compares, and validates deployment configurations across application releases, vastly reducing manual troubleshooting time.
- Engineered a feature on the tool to **concurrently** execute commands across multiple **k8s** pods rather than manually.
- Pitched the automation tool to leadership, transitioning it from an independent intern project into an **officially approved**, open-source internal Wells Fargo application with ongoing team contributions.
- Supported **CI/CD** pipelines using **GitHub Actions**, **JFrog**, and **Harness** to deploy applications into **OpenShift Container Platform (OCP)** environments via **k8s** and validated deployments through **AppDynamics** and **Grafana**.
- Improved application observability through a universal **Splunk** dashboard to replace siloed monitoring of systems.

PROJECTS

Decentralized Peer to Peer System | Notre Dame, IN January 2025 – May 2025

- Engineered a **decentralized system** where peers function simultaneously as **TCP servers** with **epoll** for socket I/O and **clients**, enabling **bidirectional** data exchange and state **synchronization** across a **distributed network**.
- Implemented a **UDP** registration protocol with a **name server** to manage membership and allow for peer discovery.
- Developed a **distributed hash table** per peer to track files in disk along with **checkpoint-log** for failure recovery.
- Designed an **eventual consistency** file-sharing model through neighbor polling and exponential backoff for failures.

Decentralized Peer to Peer System | Notre Dame, IN January 2025 – May 2025

- Created a **centralized job scheduling system** using **raw TCP sockets** and **epoll** without additional libraries.
- Architected a **fault-tolerant** job lifecycle where a coordinator node distributes zipped execution directories from clients to **stateless** workers, incorporating state-tracking to reschedule orphaned jobs if a worker node crashes.
- Developed a custom **JSON-based** communication **protocol** featuring manual socket handling, **packet acknowledgements**, and **backoff retries** to guarantee reliable data exchange even during network failures.
- Achieved a **6x** performance speedup for long-running workloads at maximum capacity of **128** concurrent jobs compared to local execution by successfully distributing the compute load across multiple machines.

LEADERSHIP

McInerney Fellows Program | Notre Dame, IN

Board Member August 2024 – Present

- **Mentored** first-year Computer Science students and proposed actionable strategies to support **FGLI** students.

Society of Latino Engineers and Scientists | Notre Dame, IN

Professional Development Chair (August 2025 – May 2026) August 2024 – Present

- Organized **3** events for **30+** students each semester, hosting industry speakers and providing career readiness support.

Data Club of Notre Dame | Notre Dame, IN

Co-President (August 2025 – Present) August 2023 – Present

- Restructured the club by strengthening project accountability, building a motivated team culture, and **creating** an **official website**, resulting in growth from **4** to **9** projects and **30** to **100** members.